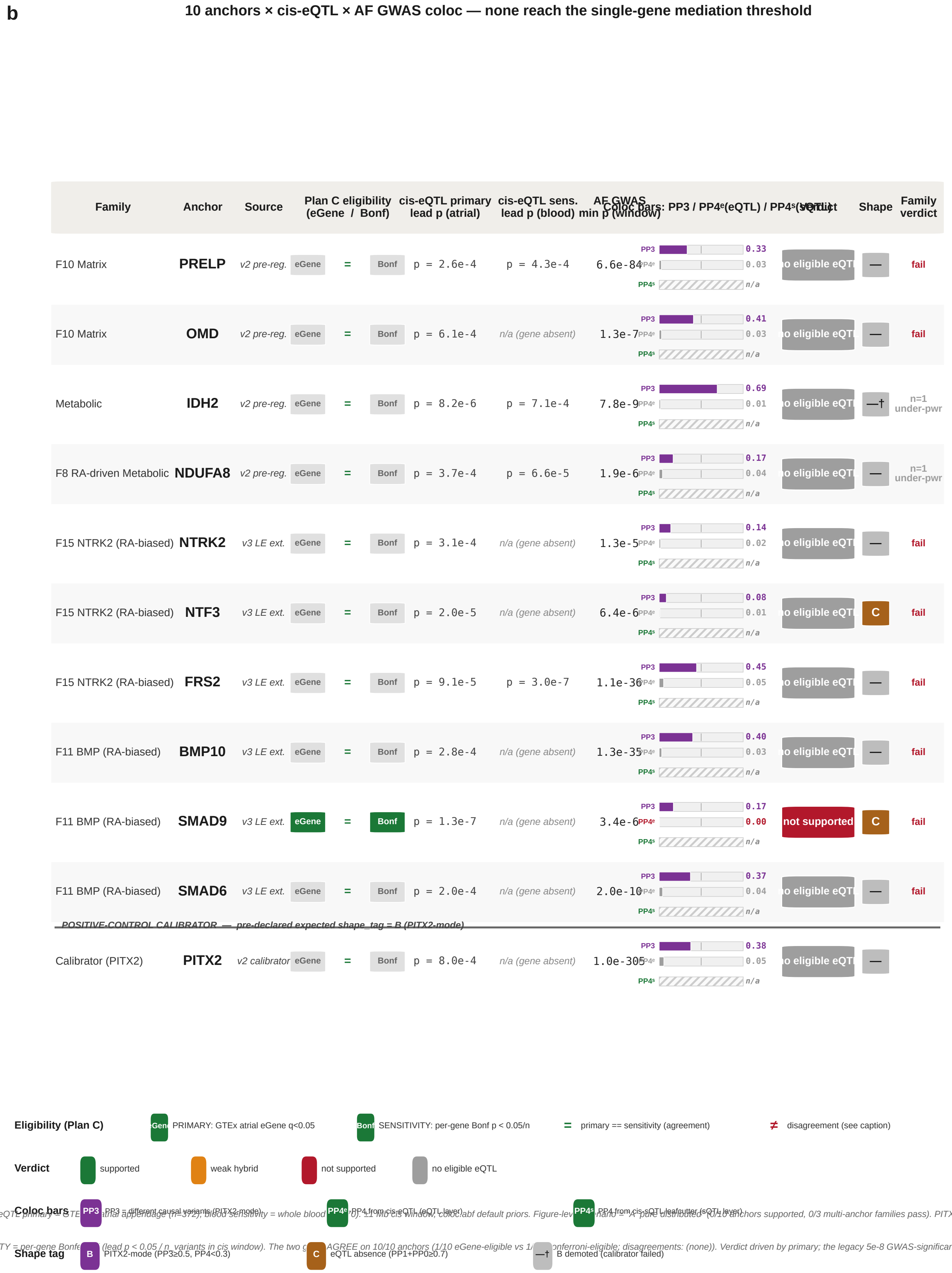

b

10 anchors × cis-eQTL × AF GWAS coloc — none reach the single-gene mediation threshold

Family	Anchor	Source	Plan C eligibility (eGene / Bonf)	cis-eQTL primary lead p (atrial)	cis-eQTL sens. lead p (blood)	AF GWAS min p (window)	Coloc bars: PP3 / PP4^e(eQTL) / PP4^s(sQTL)	Verdict	Shape	Family verdict
F10 Matrix	**PRELP**	*v2 pre-reg.*	eGene = Bonf	p = 2.6e-4	p = 4.3e-4	6.6e-84	PP3 0.33 PP4^e 0.03 PP4^s *n/a*	o eligible eQT	—	fail
F10 Matrix	**OMD**	*v2 pre-reg.*	eGene = Bonf	p = 6.1e-4	*n/a (gene absent)*	1.3e-7	PP3 0.41 PP4^e 0.03 PP4^s *n/a*	o eligible eQT	—	fail
Metabolic	**IDH2**	*v2 pre-reg.*	eGene = Bonf	p = 8.2e-6	p = 7.1e-4	7.8e-9	PP3 0.69 PP4^e 0.01 PP4^s *n/a*	o eligible eQT	—†	n=1 under-pwr
F8 RA-driven Metabolic	**NDUFA8**	*v2 pre-reg.*	eGene = Bonf	p = 3.7e-4	p = 6.6e-5	1.9e-6	PP3 0.17 PP4^e 0.04 PP4^s *n/a*	o eligible eQT	—	n=1 under-pwr
F15 NTRK2 (RA-biased)	**NTRK2**	*v3 LE ext.*	eGene = Bonf	p = 3.1e-4	*n/a (gene absent)*	1.3e-5	PP3 0.14 PP4^e 0.02 PP4^s *n/a*	o eligible eQT	—	fail
F15 NTRK2 (RA-biased)	**NTF3**	*v3 LE ext.*	eGene = Bonf	p = 2.0e-5	*n/a (gene absent)*	6.4e-6	PP3 0.08 PP4^e 0.01 PP4^s *n/a*	o eligible eQT	C	fail
F15 NTRK2 (RA-biased)	**FRS2**	*v3 LE ext.*	eGene = Bonf	p = 9.1e-5	p = 3.0e-7	1.1e-36	PP3 0.45 PP4^e 0.05 PP4^s *n/a*	o eligible eQT	—	fail
F11 BMP (RA-biased)	**BMP10**	*v3 LE ext.*	eGene = Bonf	p = 2.8e-4	*n/a (gene absent)*	1.3e-35	PP3 0.40 PP4^e 0.03 PP4^s *n/a*	o eligible eQT	—	fail
F11 BMP (RA-biased)	**SMAD9**	*v3 LE ext.*	eGene = Bonf	p = 1.3e-7	*n/a (gene absent)*	3.4e-6	PP3 0.17 PP4^e 0.00 PP4^s *n/a*	not supported	C	fail
F11 BMP (RA-biased)	**SMAD6**	*v3 LE ext.*	eGene = Bonf	p = 2.0e-4	*n/a (gene absent)*	2.0e-10	PP3 0.37 PP4^e 0.04 PP4^s *n/a*	o eligible eQT	—	fail
POSITIVE-CONTROL CALIBRATOR — pre-declared expected shape_tag = B (PITX2-mode)										
Calibrator (PITX2)	**PITX2**	*v2 calibrator*	eGene = Bonf	p = 8.0e-4	*n/a (gene absent)*	1.0e-305	PP3 0.38 PP4^e 0.05 PP4^s *n/a*	o eligible eQT	—	

Eligibility (Plan C)
- eGene PRIMARY: GTEx atrial eGene q<0.05
- Bonf SENSITIVITY: per-gene Bonf p < 0.05/n
- = primary == sensitivity (agreement)
- ≠ disagreement (see caption)

Verdict
- 🟩 supported
- 🟧 weak hybrid
- 🟥 not supported
- ⬜ no eligible eQTL

Coloc bars
- PP3 PP3 = different causal variants (PITX2-mode)
- PP4^e PP4 from cis-eQTL (eQTL layer)
- PP4^s PP4 from cis-sQTL leafcutter (sQTL layer)

Shape tag
- B PITX2-mode (PP3≥0.5, PP4<0.3)
- C eQTL absence (PP1+PP0≥0.7)
- —† B demoted (calibrator failed)

AF GWAS = GCST90624411 (Miyazawa 2023, ~1.6M); eQTL primary = GTE atrial appendage (n=372); blood sensitivity = whole blood 70). ±1 Mb cis window, coloc.abf default priors. Figure-lev nario = "A" pure distributed" (0/10 anchors supported, 0/3 multi-anchor families pass). PITX2 calibrator: shape_tag = fail-other (expected B; calibrator FAIL).

Eligibility (Plan C hybrid): PRIMARY = GTEx v8 atrial appendage eGene status (q < 0.05, BH-FDR on permutation-adjusted p_beta); SENSITIVITY = per-gene Bonfe (lead p < 0.05 / n_variants in cis window). The two g AGREE on 10/10 anchors (1/10 eGene-eligible vs 1/ onferroni-eligible; disagreements: (none)). Verdict driven by primary; the legacy 5e-8 GWAS-significance threshold is methodologically misapplied to cis-eQTL discovery and is reported only as `gate_5e8_legacy` for transparency — never as a gate in v3.

3-bar coloc: PP3 (different causal variants), PP4^e from cis-eQTL, PP4^s from cis-sQTL (leafcutter, atrial appendage). sQTL layer is structurally inapplicable: only 0/10 anchors are atrial sGenes (q<0.05; spec §10.14), so the PP4^s bar is hatched n/a for the rest. Combined RNA-layer verdict: 0/10 anchors supported across either layer. Among failed anchors, n_B = 0 reproduce PITX2-mode (B; demoted post-calibrator-fail), n_C = 2 show eQTL absence; the remaining 8 are intermediate fail-other.

Both v3 extension families (F11 BMP, F15 NTRK2/TrkB) are RA-biased in the healthy chamber prior (Fig 1f); positive coloc here would mark RA-side individual drivers, complementing the LA-side ECM (F10/PRELP/OMD) v2 set. This LA-RA symmetry across anchor cohorts strengthens the distributed-substrate interpretation.